Wiring and Interfaces Guide

Custom wiring reference for the completed build: connector strategy, UART assignments, power path, receiver wiring, GPS integration, and optional telemetry wiring.

Wiring design goals

- Keep the wiring readable enough that a second builder can service the quad without reverse-engineering the stack.
- Separate power from signal wherever practical: battery/ESC current paths should use appropriate wire gauge and physical strain relief, while UART/GPS/receiver leads can use smaller loom-style wires.
- Route the serial links so each subsystem has a clear connector destination: receiver → FC UART, GPS → FC GPS/UART, telemetry bridge → FC telemetry UART.

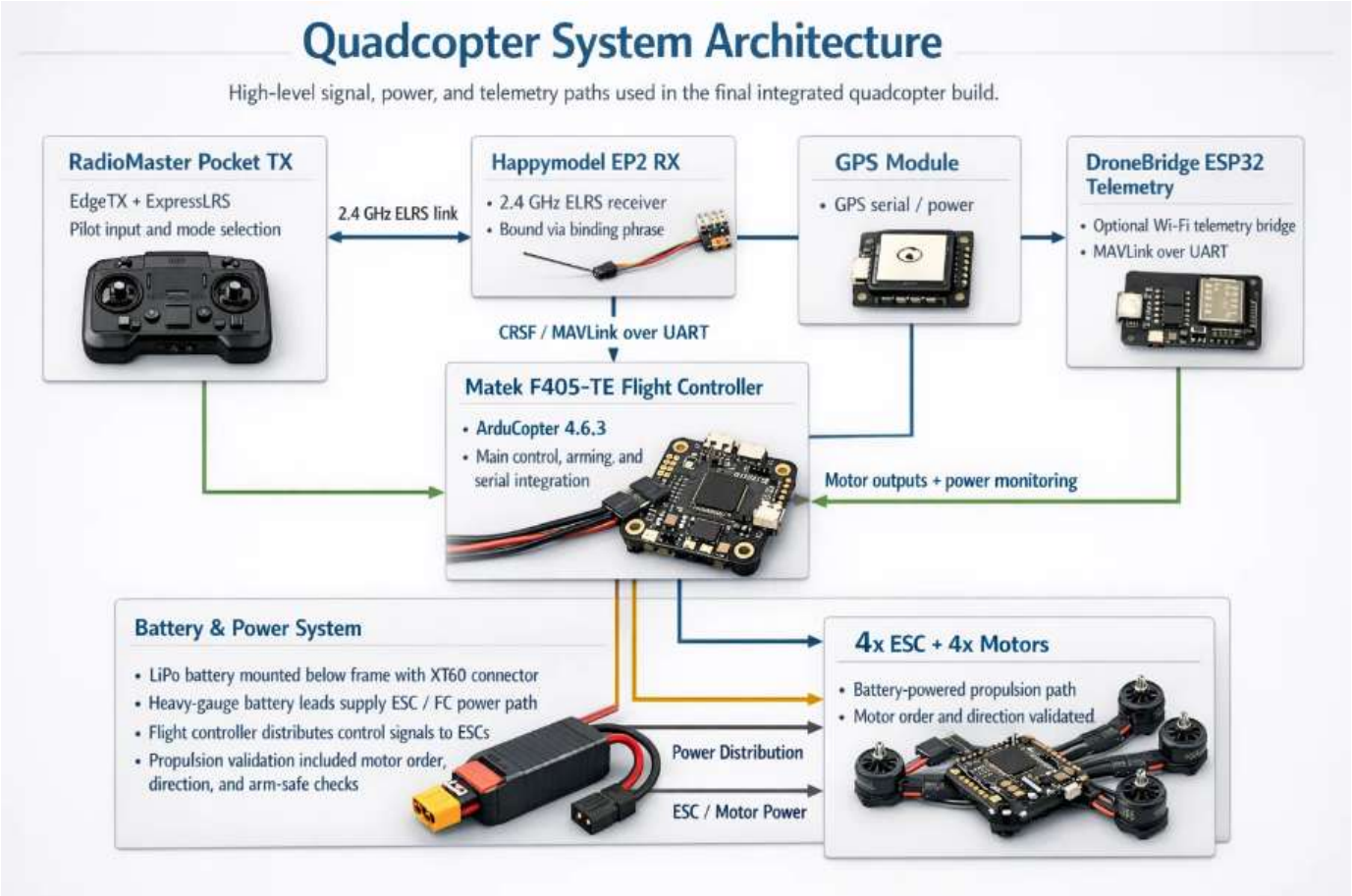

Figure 1. System architecture used to organize the control, navigation, telemetry, and propulsion paths.

Connector and loom planning

Subsystem	Typical loom / connector approach	Build note
GPS	1×6 harness with 1 mm to Dupont transition	GPS is easiest to service when the mast-side connector is accessible and the loom is strain-relieved.
ExpressLRS receiver	1×4 harness from flight-controller to receiver	Small lightweight serial / power loom; route it away from the propeller arc and secure it to reduce vibration stress.
Telemetry ESP32	1×4 harness from flight-controller telemetry UART to ESP32	Keep enough slack for USB access when flashing or reconfiguring the ESP32 module.
Camera / VTX (if used)	1×2 or small signal harnesses as needed	Keep analog / video leads distinct from motor power wiring to reduce noise coupling.
Battery and ESC power	Direct-soldered heavy-gauge wire path	Inspect polarity, solder wetting, and mechanical strain relief before powering the system.

ELRS receiver wiring

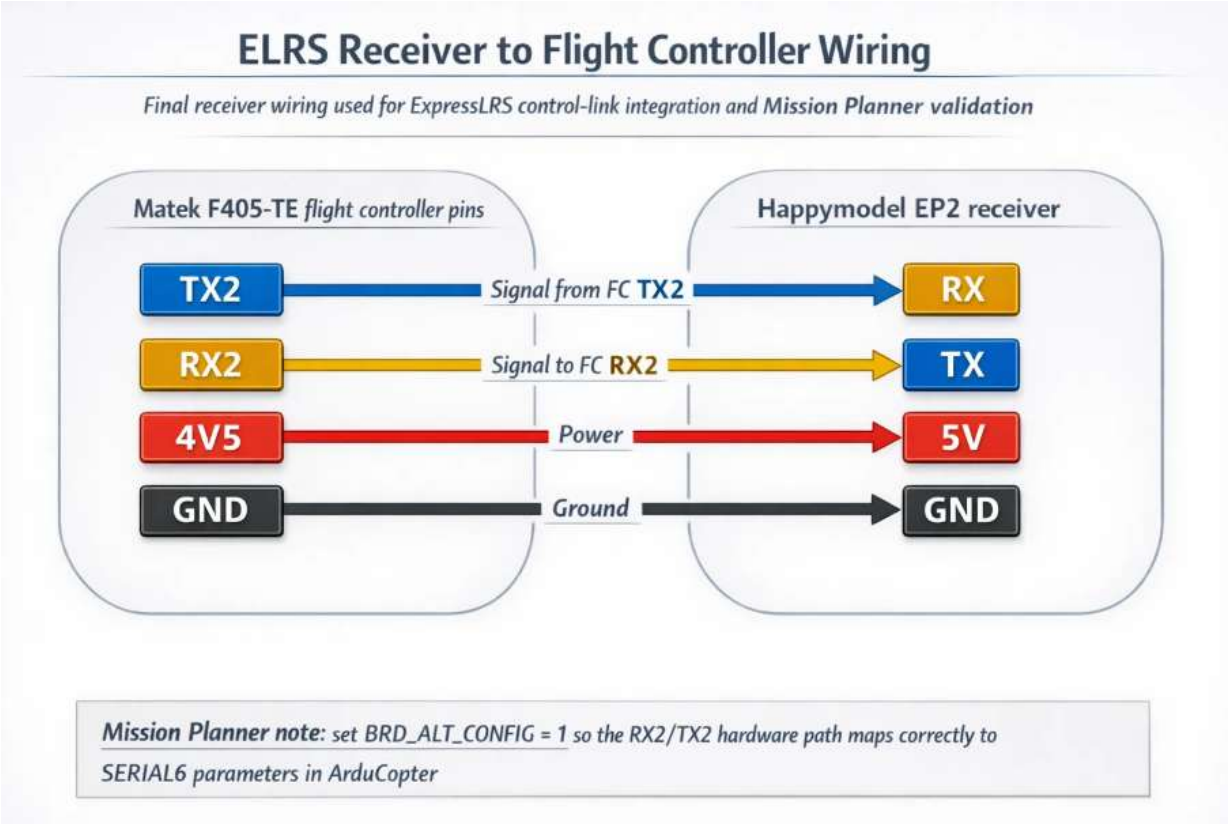

Figure 2. Receiver wiring used for the final control-link integration.

- Board-side pins: TX2, RX2, 4V5, and GND were used for the receiver wiring path.
- Cross-over rule: FC TX2 connects to receiver RX, and FC RX2 connects to receiver TX. Power and ground connect straight through.
- ArduPilot board setting: set BRD_ALT_CONFIG = 1 so the RX2/TX2 hardware path is handled correctly through the board's internal remap and the SERIAL6 parameters.
- Common protocol setting: keep RC_PROTOCOLS = 1 so ArduPilot can auto-detect the RC input protocol used by the ELRS chain.

Receiver protocol quick reference

Use case	TX link mode	RX serial protocol	ArduPilot serial protocol	RSSI_TYPE
Control only (standard ELRS / CRSF)	NORMAL	CRSF	SERIAL6_PROTOCOL = 23	3
Control + MAVLink over ELRS	MAVLINK (+ Backpack on TX)	MAVLINK	SERIAL6_PROTOCOL = 2	5

DroneBridge ESP32 telemetry wiring

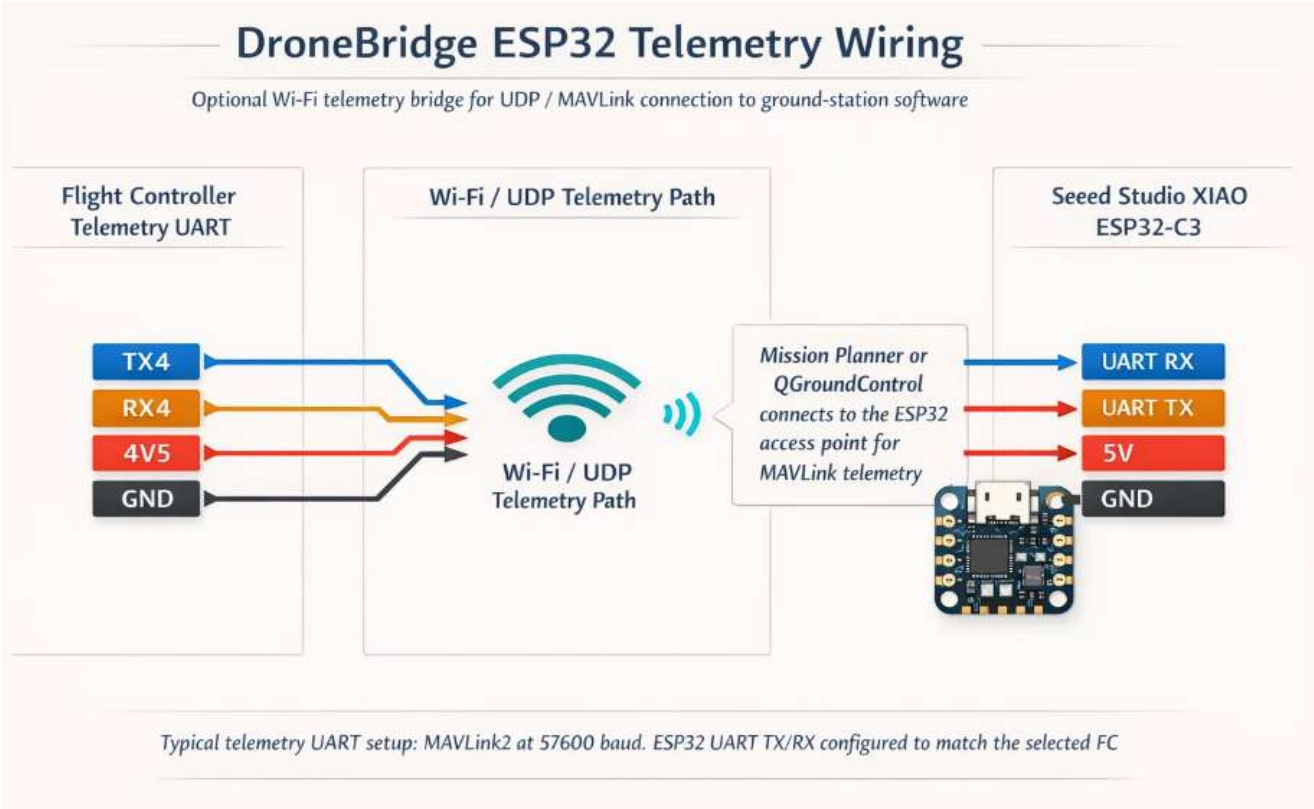

Figure 3. Optional telemetry bridge using an ESP32-C3 running DroneBridge.

- Flight-controller side: use a spare telemetry UART, commonly TX4 / RX4 plus 4V5 and GND.
- ESP32 side: wire the FC telemetry UART to the ESP32 UART pins selected in the DroneBridge config (the documented setup used the XIAO ESP32-C3 with D2/D3 and 5V/GND).
- Mission Planner side: configure the selected FC serial port for MAVLink2 at 57600 baud, then connect to the ESP32 access point over UDP/TCP depending on the ground-station workflow.

Physical wiring checks before power-up

- Battery polarity verified at the XT60 and FC power pads.
- No exposed conductor left near the carbon frame or adjacent pads after soldering.
- Motor leads restrained and routed so propeller clearance is preserved under vibration.
- Receiver, GPS, and telemetry wires bundled lightly — tight enough to avoid prop contact, loose enough to avoid stressing solder joints.
- USB / service access maintained for the flight controller, radio, and ESP32 so firmware and parameter updates are still practical after final assembly.